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Lecture 09A

Before-Tutorial-Session

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Example. Consider $B = \begin{bmatrix} 101 & -1 \\ 2 & 104 \end{bmatrix}$. Observe that

$$\begin{bmatrix} 101 & -1 \\ 2 & 104 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 102 \\ -102 \end{bmatrix} = 102 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

So, what can we say?

Recall: if $\hat{A} \underline{v} = \lambda \underline{v}$
then we say

$\therefore 102$ eigenvalue

$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ eigenvector

- $\underline{v}$ is an eigenvector with corr. eigenvalue λ
- $(\lambda, \underline{v})$ is an eigenvalue - eigenvector pair

Let's do these questions today. Tut07, Q1+Q9

Question 1. Let $A = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$. *← very routine*

- (a) Find the eigenvalues and eigenvectors of A .
- (b) Verify that the trace equals the sum of the eigenvalues, and the determinant equals their product.
- (c) Consider $B = A + 100I = \begin{bmatrix} 101 & -1 \\ 2 & 104 \end{bmatrix}$. Find the eigenvalues and eigenvectors of A .

Question 9. *← less routine*

- (a) A square matrix A is invertible if and only if $\lambda = 0$ is not an eigenvalue of A .
- (b) Suppose A is an invertible matrix. If λ is an eigenvalue of A and v is a corresponding eigenvector, then $\lambda^{-1} = \frac{1}{\lambda}$ is an eigenvalue of the inverse A^{-1} , and v is a corresponding eigenvector.

Question 1. Let $A = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$.

(a) Find the eigenvalues and eigenvectors of A .

What is the characteristic polynomial of A ?

$$\begin{aligned} 1. \quad \det(A) & \quad \det \begin{bmatrix} 1-\lambda & -1 \\ 2 & 4-\lambda \end{bmatrix} \\ \checkmark 2. \quad \det(A - \lambda I) & = (1-\lambda)(4-\lambda) - (-1)(2) \\ 3. \quad \det(A + \lambda I) & = 6 - 5\lambda + \lambda^2 \\ & = (\lambda - 2)(\lambda - 3) \end{aligned}$$

$\therefore \lambda = 2, \lambda = 3$ are the eigenvalues,

Question 1. Let $A = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$.

(a) Find the eigenvalues and eigenvectors of A .

Fix $\lambda = 3$. What is the corresponding eigenspace E_3 ?

1. $\text{null}(A)$

$A - \lambda I$

2. $\text{null}(A - 3I)$

3. $\text{null}(A + 3I)$

4. $\text{null}(A - 2I)$

5. $\text{null}(A + 2I)$

i.e. solve $\begin{bmatrix} -2 & -1 \\ 2 & 1 \end{bmatrix} \underline{x} = 0$

i.e. $E_3 = \left\{ s \begin{bmatrix} 1 \\ -2 \end{bmatrix} : s \in \mathbb{R} \right\}$
 $\hookrightarrow$ eigenspace corr. to $\lambda = 3$

Setting $\det(\hat{A} - \lambda \hat{I}) = 0$
 forces $\text{null}(\hat{A} - \lambda \hat{I})$
 to be non-trivial

not sure if λ or x

Question 1. Let $A = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$.

(a) Find the eigenvalues and eigenvectors of A .

Fix $\lambda = 2$. What is a corresponding eigenvector?

$$E_2 = \text{null}(A - 2I)$$

$$\text{solve } \begin{bmatrix} -1 & -1 \\ 2 & 2 \end{bmatrix} \vec{x} = \vec{0}$$

$$\text{So, } E_2 = \left\{ s \begin{bmatrix} -1 \\ 1 \end{bmatrix} : s \in \mathbb{R} \right\}$$

~~$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$~~ $\rightarrow$ cannot be zero vector

$\checkmark \begin{bmatrix} -1 \\ 1 \end{bmatrix} \rightarrow$ belongs to E_2 if set $s = 1$

$\checkmark \begin{bmatrix} 1 \\ -1 \end{bmatrix} \rightarrow$ belongs to E_2 if set $s = -1$

$\checkmark \begin{bmatrix} 1048576 \\ -1048576 \end{bmatrix} \rightarrow$ belongs to E_2 if $s = 1048576$

Question 1. Let $A = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$.

(a) Find the eigenvalues and eigenvectors of A .

Fix $\lambda = 2$. Which of the following will be accepted as a correct answer to Q1(a)?

All are correct

Statement A: The eigenspace $E_2 = \{ s \begin{bmatrix} 1 \\ -1 \end{bmatrix} : s \in \mathbb{R} \}$.

Statement B: The eigenspace $E_2 = \text{span}\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \}$.

Statement C: One corresponding eigenvector is $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Statement D: One corresponding eigenvector is $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

Question 1. Let $A = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$.

(b) Verify that the trace equals the sum of the eigenvalues, and the determinant equals their product.

Let $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. What is $\text{tr}(M)$ and $\det(M)$?

✓ 1. $\text{tr}(M) = a + d$ and $\det(M) = ad - bc$

2. $\text{tr}(M) = ad - bc$ and $\det(M) = a + d$

$$\lambda_1 = 3, \lambda_2 = 2$$

$$\text{tr}(A) = 1 + 4 = 5$$

$$\leftarrow \lambda_1 + \lambda_2 = 5$$

$$\det(A) = 4 \times 1 - (-1)(2) = 6$$

$$\leftarrow \lambda_1 \lambda_2 = 6$$

} Proof Tut 7, Q10.

Question 1. Let $A = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$.

(c) Consider $B = A + 100I = \begin{bmatrix} 101 & -1 \\ 2 & 104 \end{bmatrix}$. Find the eigenvalues and eigenvectors of B .

Solution:

- Option 1: Repeat as in (a)
- Option 2: Guess the eigenvectors of B .

For $\hat{A}$, $\lambda_1 = 3$, $v_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$
 $\lambda_2 = 2$, $v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Guess! $\hat{B} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = 103 \begin{bmatrix} 1 \\ -2 \end{bmatrix}$
 $\downarrow$
For $\hat{B}$, $\lambda_1 = 103$, $v_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$
 $\lambda_2 = 102$, $v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Question 1. Let $A = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$.

(c) Consider $B = A + 100I = \begin{bmatrix} 101 & -1 \\ 2 & 104 \end{bmatrix}$. Find the eigenvalues and eigenvectors of B .

Solution: Let (λ, v) be an eigenvalue-eigenvector pair of A .

$$\hat{A}v = \lambda v$$

$$\text{set } \hat{B} = \hat{A} + 100\hat{I}$$

$$\hat{B}v = (\hat{A} + 100\hat{I})v = \hat{A}v + 100\hat{I}v$$

$$= \hat{A}v + 100v$$

$$= \lambda v + 100v = (\lambda + 100)v$$

Then **BLANK** is an eigenvalue-eigenvector pair of B .

$$(\lambda + 100, v)$$

Question 9.

(a) A square matrix A is invertible if and only if $\lambda = 0$ is not an eigenvalue of A .

What is the contrapositive of the statement?

$(p \text{ if and only if } q)$ is equivalent to $(\neg p \text{ if and only if } \neg q)$

$\hat{A}$ is NOT invertible if and only if 0 is an eigenvalue of $\hat{A}$

Question 9.

(a) A square matrix A is invertible if and only if $\lambda = 0$ is not an eigenvalue of A .

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(contrapositive) $\hat{A}$ is not invertible iff 0 is an eigenvalue of $\hat{A}$

Solution: ($\Leftarrow$) Suppose A has eigenvalue 0. Then

$v \in \text{null space of } \hat{A}$

with eigenvector v

$$\hat{A}v = \lambda v = 0 \cdot v = 0$$

BLANK

$$\rightarrow \hat{A}v = 0v = 0$$

So, the null space of A is non-trivial. Therefore, A is not invertible.

Question 9.

(a) A square matrix A is invertible if and only if $\lambda = 0$ is not an eigenvalue of A .

HIDDEN

↳ (Contrapositive) A is not invertible iff 0 is an eigenvalue of A .

Solution: ($\Rightarrow$) Suppose A is not invertible.

$\Rightarrow$ nullspace of A is nontrivial

$\Rightarrow$ there is a non-zero x s.t. $A \cdot x = \underline{0} = 0 \cdot \underline{v}$

$\nearrow \lambda$ eigenvalue
 $\nwarrow$ eigen vector

Then

BLANK

is an eigenvalue-eigenvector pair of A .

Question 9.

- (b) Suppose A is an invertible matrix. If λ is an eigenvalue of A and v is a corresponding eigenvector, then $\lambda^{-1} = \frac{1}{\lambda}$ is an eigenvalue of the inverse A^{-1} , and v is a corresponding eigenvector.

Albert's Solution: This prove is not correct

Let $A = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$. Then $v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ is an eigenvector with eigenvalue $\lambda = 2$.

So, $A \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Hence, $\frac{1}{2} \left(A \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

So, $A \left(\frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

So, $\frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = A^{-1} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Therefore, $\frac{1}{\lambda}$ is an eigenvalue of A^{-1} with corresponding eigenvector v .

Question 9.

(b) Suppose A is an invertible matrix. If λ is an eigenvalue of A and v is a corresponding eigenvector, then $\lambda^{-1} = \frac{1}{\lambda}$ is an eigenvalue of the inverse A^{-1} , and v is a corresponding eigenvector.

Proof

Since (λ, v) is an eigenvalue-eigenvector pair of A , $Av = \lambda v$

because $\lambda \neq 0$

↑ (Q91a)
because A is
invertible

$$\Rightarrow \frac{1}{\lambda} Av = v$$

$$\Rightarrow A\left(\frac{1}{\lambda} v\right) = v$$

$$\Rightarrow \frac{1}{\lambda} v = A^{-1} v$$

$\therefore \left(\frac{1}{\lambda}, v\right)$ is an eigenvalue-eigenvector pair of A^{-1} .